

Project Epsilon 3

Condensed Professional Project Report

Application Type	Role-based trekking and expedition management web application
Technology Stack	Flask, Jinja2, SQLAlchemy, SQLite, HTML5, Custom CSS
Main Goal	Manage expeditions, guide assignments, and trekker bookings through separate dashboards

Executive Summary

Project Epsilon 3 is a compact but complete web platform for managing trekking expeditions. It supports three distinct roles: manager, guide, and trekker. The design uses Flask blueprints for modular routing, SQLAlchemy models for persistence, and Jinja2 templates for role-specific dashboards.

The application demonstrates the full lifecycle of a booking platform: account creation, role verification, expedition publishing, guide assignment, seat tracking, and reservation recording.

Core Objectives

- Provide separate interfaces for administration, expedition operations, and customer booking.
- Store expedition, user, and booking data in a normalized relational structure.
- Allow managers to verify guides and suspend non-admin accounts.
- Keep expedition seat availability synchronized with bookings and guide updates.

Quick Metric	Value
Primary roles	3
Core tables	3
Controller modules	3
Seed expeditions	4